How to Reverse Engineer Any Institutional Trading System Using REA to Build a One-Person Hedge Fund From Scratch

An Evidence-Driven Framework for Software Reconstruction, Quantitative Strategy Research, and Autonomous Trading Infrastructure

***Abstract*—This paper presents a practical framework for using REA (Reverse Engineer Anything) and its configured Model Context Protocol (MCP) tools, to investigate authorized trading software, reconstruct observable mechanisms, and build an independent quantitative research desk. The proposed system combines software investigation, market-data ingestion, hypothesis generation, strategy implementation, mation, cost-aware backtesting, risk controls, and continuous evaluation. Rather than assuming that a hedge fund's proprietary strategy can be recovered from public observations, the method distinguishes verified implementation evidence from inferred decication rules. Each reconstructed mechanism becomes a falsifiable hypothesis, each strategy must survive out-of-sample validation, and every deployment decision is governed by explicit risk limits. The objective is not to promise access to confidential strategies or guaranteed trading profits, but to turn reverse engineering into a reproducible research process capable of producing independently testable quantitative models.**

***Index Terms*—REA, quantitative finance, institutional trading, reverse engineering, MCP, hedge fund infrastructure, market microstructure, strategy discovery, execution modeling, backtest validation, systematic trading, risk management.**

I. From Reverse Engineering to a Quant Desk

Institutional trading systems combine data ingestion, pricing models, order management, execution logic, portfolio constraints, and risk controls. Their visible outputs—quotes, orders, cancellations, fills, and price movements—provide evidence about external behavior but rarely reveal a unique internal strategy. Different implementations may generate similar observations, and apparent historical profitability can arise from noise, unrealistic execution assumptions, or repeated model selection.

REA provides a structured investigation workflow for supported software artifacts and observable behavior. Through the tools actually available in its configured environment, a coding agent can investigate authorized targets, trace supported code paths, strats, organize evidence, and help construct an independent implementation. Thoscontrurs an independent implementation. The research must record what was directly observed, what was inferred, and what remains unknown. Private source code, credentials, or proprietary strategies are not assumed to be accessible.

A. The One-Person Research Desk

The proposed desk integrates six capabilities: an evidence collection layer, a market-data engine, a model and strategy laboratory, an independent validation engine, a risk-cont layer, an independent validation engine, a risk-control layer, and an experiment ledger. The researcher specifies objectives and approves changes, while automation handles repeatable analysis, testing, reporting, and monitoring. These components form a controlled research loop in which every hypothesis has a reproducible implementation, explicit assumptions, and measurable rejection criteria.

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TABLE I
From Trading Idea to Research Desk

Component	Informal Workflow	Proposed Desk
Investigation	Visual assumptions	Evidence ledger
Strategy	Manual experimentation	Versioned models
Validation	Best historical result	Holdout testing
Execution	Idealized fills	Explicit cost model
Risk	Discretionary limits	Independent controls

B. From Software Evidence to Economic Value

A reconstructed mechanism becomes useful only when its conserquences can be measured. Let R_g denote gross strategy return and C_f, C_s, C_{m}, and C_o represent fees, slippage, market impact, and other operating costs on a consistent return basis. Equation (1) defines the net-return quantity used to evaluate candidate strategies.

$$R_{\mathrm{net}} = R_{\mathrm{g}} - C_f - C_s - C_{\mathrm{m}} - C_{\mathrm{o}} \tag{1}$$

Positive historical net returns are not sufficient evidence of a durable edge. Candidate strategies must be evaluated using chronological holdouts, realistic execution assumptions, multiple-testing controls, and stress scenarios. A reconstructed software feature may improve a simulator without creating an exploitable trading opportunity.

C. The End-to-End REA Research Workflow

Figure 1 summarizes the proposed pipeline from authorized investtigation to independent strategy evaluation. Failed tests return the investigation to hypothesis formation; candidates that pass proceed to controlled paper trading and further risk review.

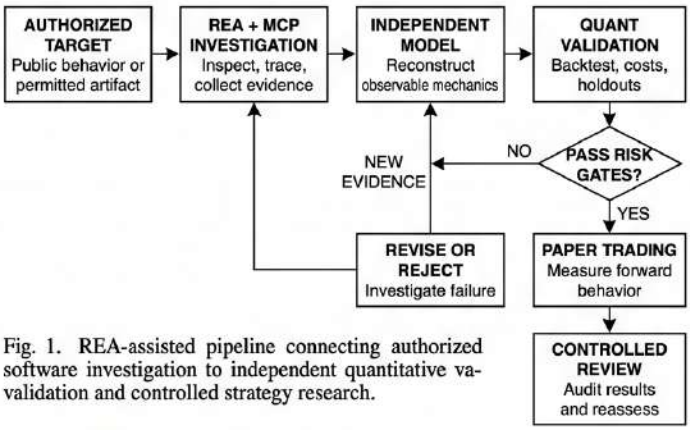


Fig. 1. REA-assisted pipeline connecting authorized software investigation to independent quantitative va-validation and controlled strategy research.

D. What Makes the System Institutional-Grade?

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Institutional-grade engineering is defined by reliability and controls, not interface complexity. Market data must have validated timestamps and provenance. Strategy code must be versioned and reproducible. Backtests must distinguish signals from executable fills. Exposure limits must operate independently of strategy logic. Finally, every experiment—including failed candidates—must be recorded to reduce selection bias and permit independent review.

E. Research Scope and Contributions

This paper develops the investigation architecture, quantitative reconstruction methodology, feature and hypothesis design, backtest validation protocol, and deployment controls required to build a one-person quant research desk. Section II examines the evidence model and technical architecture. Section III develops observable-event features and competing hypotheses. Section IV establishes statistical and execution validation. Section V specifies risk controls and paper-trading progression. Section VI addresses reproducibility, limitations, and conclusions. The resulting framework treats strategy discovery as a scientific process: collect evidence, test explanations, reject unsupported models, and retain only conclusions justified by the data.

II. The REA-Assisted Reconstruction Architecture

A systematic reverse-engineering investigation separates three tasks: observing a trading system, reconstructing a plausible explanation of its behavior, and testing whether that explanation has quantitative value. REA supplies the investigation interface through its configured CLI and MCP workflows. The coding agent interprets evidence and builds an independent model, while quantitative tools test its predictions. These functions must remain distinct. Recovered code does not automatically establish a trading strategy, and matching historical behavior does not prove that the underlying implementation has been uniquely identified.

A. Separating Observable Behavior From Hidden Logic

Three evidence levels must remain separate. Let x_t denote the observable market state, s_t the system's potentially hidden internal state, y_t the observed trading action, and θ the unknown decision parameters. An abstract state-space representation is given by Equation (1).

$$y_t = F_\theta(x_t, s_t) + \varepsilon_t \tag{1}$$

Here, F_θ represents the unknown decision mechanism and ε_t captures observation error, timing uncertainty, or unmodeled effects. The observed action may be a quote update, order submission, cancellation, or fill. Because inventory, private signals, execution constraints, and internal state transitions may remain unobserved, several candidate models can explain the same s outcome. The objective is therefore to identify mechanisms consistent with the evidence, not to claim that one observed sequence reveals an institution's complete proprietary strategy.

B. From Evidence to Model Selection

REA organizes supported investigation around three stages: decompile, understand, and recreate. Depending on the artifact and available analysis provider, the agent may inspect readable code, strings, symbols, pseudocode, function relationships, or observable application behavior. It then traces the relevant logic and reconstructs an independent imimentation. The analysis must preserve the distinction between extracted evidence and an inferred explanation. Equation (2) defines a model-selection objective for comparing competing reconstructions.

$$J(M) = E_{\text{holdout}}(M) + \lambda C(M) + \mu U(M) \tag{2}$$

In Equation (2), E_{holdout} is the error on observations withheld from model construction, $C(M)$ penalizes unnecessary complexity, and $U(M)$ measures unsupported assumptions. The nonnegative weights λ and μ are chosen before evaluating candidates. Lower values favor models that explain independent observations with fewer unsupported assumptions. This objective is a research criterion, not a profitability score; its components and normalization must be defined for the specific investigation.

C. REA as the Investigation Interface

REA connects compatible coding agents to reverse-engineering tools through CLI and MCP workflows. The first operation is a capability audit: inspect the repository documentation, available MCP tool schemas, installed analysis providers, target format, and permitted access. Native binaries may require a supported tool such as Hopper or Ghidra, while some JavaScript investigations need no native decompiler. The agent must verify actual availability before selecting a workflow. If an artifact cannot be analyzed, it records the limitation and uses documented behavior or other authorized evidence instead [1].

D. Quantitative Hypotheses and Independent Validation

Once a mechanism has been reconstructed, translate it into falsifiable hypotheses. A price-rounding hypothesis can be checked against valid tick sizes; an execution hypothesis can be evaluated against timeanped submissions and fills; an inventory-control hypothesis can be compared with observed quote skew as positions change. Use chronological data, explicit transaction costs, and a separate validation period. If multiple strategies or parameter settings are tried, record the search process and account for selection bias. A visually convincing match or profitable in-sample backtest is not sufficient evidence of a durable edge.

E. Ownership of the Reconstruction Pipeline

The workflow assigns each stage a clear deliverable. REA investigates authorized artifacts and preserves evidence; the researcher formulates competing explanations; an independent implementation expresses the selected model as executable code; and the validation system measures discrepancies on separate observations. Table II summarizes the responsibilities and the evidence required at each stage.

TABLE II
Responsibilities in the Reconstruction Pipeline

Stage	Primary responsibility	Required evidence
Target	Define scope and access	Target dossier
Investigation	Trace supported behavior	REA evidence records
Hypothesis	Explain observed outcomes	Falsifiable model
Reconstruction	Implement independently	Reproducible code
Validation	Test unseen observations	Error analysis
Risk review	Assess economic relevance	Costs and failure cases

F. Interpretation and Limits

The framework can reconstruct selected observable mechanisms without establishing that an institution's complete strategy has been recovered. Public market data may reveal statistical regularities but rarely identifies a unique internal decision process. Decompilation may expose useful implementation clues without reproducing original source code. Each conclusion must therefore state its evidence, assumptions, and limits before advancing to the quantitative strategy experments in the next section.

III. From Observable Events to Quantitative Models

After the investigation architecture has been established, the central research problem becomes identifying which mechanisms are consistent with the available evidence. A trading system may expose quotes, trades, order acknowledgements, cancellations, fills, and changes in displayed state. These observations can consrain a model, but they do not uniquely reveal the decision process of the institution producing them. The objective is to develop competing, falsifiable explanations and determine which survive independent tests.

The most useful investigations connect software behavior to a precise market question. If an authorized trading engine changes its quotes after inventory changes, does the adjustment correspond to an inventory-control rule? If quotes disappear before a price move, is the pattern explained by cancellation behavior, volatility, or ordinary liquidity withdrawal? If observed order-flow imbalance precedes a price change, does it predict returns after controlling for spread, depth, volatility, and trading costs? Each question requires a measurable event, a candidate mechanism, and evidence capable of rejecting the explanation.

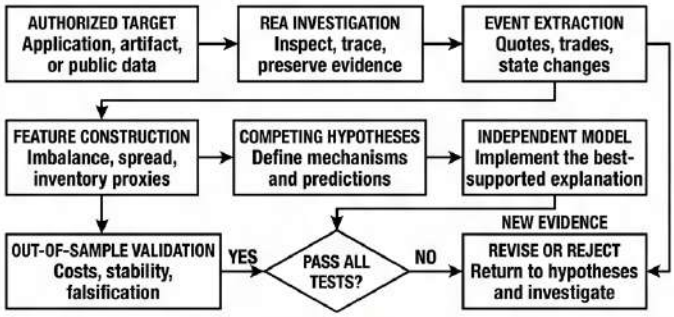


Fig. 1. Evidence-to-model workflow for reverse-engineering observable trading mechanics and validating independent quantitative hypotheses.

A. Reconstructing the Observable Event Stream

Before fitting any model, normalize the available evidence into a chronological event stream. Each record should include an event identifier, source, instrument, event type, timestamp, observed fields, and data-quality status. When accessible, preserve exchange timestamps separately from local observation timestamps to quantify timing uncertainty. Distinguish submitted orders from acknowledged orders, accepted orders from fills, and cancellations requested from cancellations confirmed. Never infer a missing event solely because a downstream state changed.

B. Constructing Market-Microstructure Features

On markets with reliable order-book data, begin with interpretable features. Let Q_t^b and Q_t^a denote displayed bid and ask quantities at the best prices. A normalized top-of-book imbalance is defined by Equation (1).

$$I_t = \frac{Q_t^b - Q_t^a}{Q_t^b + Q_t^a} \tag{1}$$

The statistic lies between minus one and one when the denominator is positive. Positive values indicate greater displayed bid quantity, not guaranteed upward movement or proof of informed buying. Where event-level quote data exists, construct order-flow imbalance from changes in best-bid and best-ask prices and displayed quantities. Cont, Kukanov, and Stoikov find that short-horizon price changes are related to order-flow imbalance, with sensitivity depending on market depth [3]. This motivates a hypothesis for testing, not an assumption of universal or profitable predictability.

C. Turning Features Into Competing Hypotheses

Define the prediction target before selecting a model. For example, let m_t denote the observed mid-price and h the prediction horizon. The forward mid-price return is $r_{t,h} = (m_{t+h} - m_t)/m_t$. Candidate hypothesis H_1 may predict this return from order-flow imbalance; H_2 may add spread, volatility, depth, and recent returns; H_3 may test whether the relation changes by instrument or market regime. Compare the candidates on the same chronological splits. Additional complexity is justifiird only when it improven reliable out-of-sample performance.

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$$r_{t,h} = \alpha + \beta \frac{\mathrm{OFI}_t}{D_t} + \gamma^T z_t + \varepsilon_{t,h} \tag{2}$$

In Equation (2), OFI_t denotes a defined order-flow imbalance measure, D_t a strictly positive depth normalization, z_t a vector vector of control variables, and $\varepsilon_{t,h}$ the residual. The exact normalization, sampling interval, and estimation procedure must be specified before testing. Coefficients estimated from historical data describe conditional association unless a separate identification design supports a causal interpretation.

D. Falsification Before Optimization

Predefine observations that would contradict each hypothesis. An inventory-skew hypothesis should fail if the proposed quote adjustment does not track the available inventory proxy after controlling for market conditions. An imbalance predictor should fail if its effect disappears on untouched periods, reverses unpredictably across regimes, or becomes negative after realistic costs. Missing data, timestamp misalignment, survivorship, and selection bias must be investigated before changing model parameters. Rejecting an attractive hypothesis is a valid research outcome.

E. Comparing Candidate Reconstructions

Table III illustrates how the same observable pattern can support multiple competing explanations. The purpose of the comparison is to design discriminating tests rather than choose the most preprofitable story after observing the results.

TABLE III
Observation-to-Hypothesis Test Design

Observation	Competing explanations	Discriminating test
Quote skew	Inventory control; adverse selection	Condition on inventory and flow
Quote withdrawal	Risk limit; volatility response	Compare volatility regimes
Short-term price impact	Informative flow; transient pressure	Test multiple forward horizons
Apparent fills	Genuine execution; optimistic simulation	Reconcile with event records

F. From Reconstruction to Economic Value

A model that explains observed mechanics is not automatically a strategy. Economic evaluation begins only after the mechanism has been specified independently and its predictions have survived holdout testing. The next stage converts these surviving hypotheses into reproducible strategy experiments with execution costs, risk constraints, multiple-testing controls, and explicit criteria for rejection. The research process must preserve negative results alongside positive findings to prevent selective reporting.

IV. The Backtest Validation Gauntlet

Reconstructing a plausible trading mechanism is only the beginning of quantitative research. The next task is to determine whether its predictions generate economically meaningful performance outside the observations that inspired the model. A strategy can appear profitable because of accidental correlations, unrealistic fills, excessive parameter search, or a favorable market regime. The validation process must therefore challenge both the reconstructed mechanism and the economic assumptions used to trade it.

A. Chronological Separation and Leakage Control

Partition the historical dataset into development, validation, and untouched test periods ordered by time. Construct features only from information available at the decision timestamp. When labels overlap across prediction horizons, purge affected observations and apply an appropriate embargo before evaluating dependent samples. Fit normalizers, thresholds, and model parameters using training data alone. Preserve the complete experiment log, including abandoned configurations, because repeated strategy selection changes the evidential value of the strongest observed result.

B. Modeling Execution Rather Than Idealized Returns

Every simulated trade must follow an explicit execution policy. A market order executes against available liquidity under a documented fill assumption; a limit order may remain unfilled, execute partially, or receive adverse selection. Include commissions, bid-ask spread, slippage, funding where applicable, and market impact when the available data supports estimating them. Candle-only backtests cannot reliably reconstruct queue priority or intrabar execution order. Unsupported execution details must be modeled conservatively and disclosed rather than invented.

Define net strategy return using Equation (3), with every cost term expressed over the same capital and time basis.

$$r_{\text{net},t} = r_{\text{gross},t} - c_{\text{fee},t} - c_{\text{slippage},t} - c_{\text{funding},t} - c_{\text{impact},t} \quad (3)$$

Costs must be applied when trades occur, not subtracted afterward using an arbitrary average. For strategies with frequent turnover, small execution errors can reverse the ranking of candidate models. Run sensitivity analysis under adverse assumptions about spreads, latency, fill rates, and market impact. If the data does not identify these costs precisely, report a range of plausible outcomes rather than a single apparently exact estimate.

C. Performance and Risk Measurement

Report net annualized Sharpe ratio, maximum drawdown, turnover, exposure, trade count, tail losses, and performance by market regime. For a stationary return series with a defined annualization factor A, the conventional annualized Sharpe ratio is given in Equation (4). Serial dependence and non-normal returns require additional care when interpreting its statistical significance.

$$\text{SR} = \sqrt{A} \cdot \frac{\text{mean}(r_{\text{net}})}{\text{sd}(r_{\text{net}})}. \quad (4)$$

Evaluate return concentration, skewness, kurtosis, drawdown duration, and sensitivity to individual trades. A high Sharpe ratio based on very few observations may be unreliable. Compare the candidate against simple baselines, including passive exposure and an elementary strategy with comparable risk. Report confidence intervals or suitable dependence-aware uncertainty estimates where feasible. Performance metrics describe the tested sample; they do not guarantee future profitability.

D. Correcting for Strategy Selection

Searching many indicators, parameters, instruments, and holding periods increases the probability of selecting a historically impressive result by chance. Bailey and López de Prado's Deflated Sharpe Ratio addresses selection bias and non-normality; probability-of-backtest-orfitting methods examine how frequently an in-sample winner performs poorly out of sample [2], [4]. Record how many variants were tested, including discarded experiments, and distinguish a prespecified hypothesis from an exploratory discovery.

For a combinatorially symmetric cross-validation analysis, let p_k denote the out-of-sample percentile rank of the strategy selected as the in-sample winner in split k, with zero representing the lowest rank and one the highest. A simplified estimate of the probability of backtest overfitting is given in Equation (5).

$$\text{PBO} = \frac{1}{K} \sum_{k=1}^{K} \mathbb{I}(p_k < 0.5) \quad (5)$$

Here, K is the number of valid splits and $\mathbb{I}$ is the indicator function. PBO estimates how often the in-sample winner ranks below the out-of-sample median under the chosen splitting procedure. It is a diagnostic, not proof that a strategy will fail, and its interpretation depends on the sampling design and dependence structure. Use it alongside untouched chronological testing, not as a replacement for it [5].

E. The Five-Gate Validation Protocol

Every candidate reconstructed from REA-assisted investigation must pass a sequence of explicit gates before promotion to paper trading. Gate 1 verifies data provenance, timestamp integrity, and leakage controls. Gate 2 tests whether the independent model reproduces the targeted behavior on evidence not used to construct it. Gate 3 evaluates net performance under realistic execution assumptions. Gate 4 tests chronological holdouts, regime stability, and as the effects of multiple trials. Gate 5 examines drawdowns, tail exposures, concentration, and operational failure scenarios. A failure at any gate returns the candidate for revision or rejection.

Table IV summarizes the decision protocol. Passing these gates qualifies a model for further prospective evaluation, not automatic deployment with live capital.

TABLE IV
The Five-Gate Strategy Validation Protocol

Gate	Validation question	Failure response
Data integrity	Are timestamps and inputs valid?	Reject dataset
Model fidelity	Does behavior match held-out evidence?	Revise model
Execution costs	Does net performance survive costs?	Reject claimed edge
Statistical robustness	Does the result survive holdouts and trial corrections?	Reduce confidence
Risk and operations	Does it survive stress and failure tests?	Block promotion

F. An Evidence-Based Promotion Decision

Preserve all failed experiments in the research record and report why each candidate was rejected. Promote only those that meet prespecified thresholds and demonstrate stable performann under independent evaluation. The surviving strategy proceeds to controlled paper trading, where simulated decisions can be compared with subsequent market events and executions can be monitored. The next section examines position sizing, portfolio exposure, and risk controls required before any consideration of live capital.

V. Risk Controls and Controlled Deployment

A reconstructed strategy that survives statistical validation still requires a separate operational assessment. Historical evidence cannot guarantee that a model will behave safely under live conditions, where network failures, stale prices, partial fills, exchange outages, and rapid market transitions can create losses absent from the backtest. The deployment objective is **not** to maximize activity. It is to ensure that every strategy operates inside explicit exposure limits, that abnormal behavior is detected promptly, and that execution can be stopped independently of the strategy logic.

A. Position Sizing Under Model Uncertainty

Let E denote current account equity, f the fraction allocated to a defined risk budget, and L the estimated loss per unit of position under a specified adverse-move scenario. A basic risk-budget sizing rule is stated in Equation (6).

$$q_{\max} = \frac{f \cdot E}{L}, \tag{6}$$

Here, $q_{\max}$ is the maximum position quantity under the assumed scenario. The loss estimate must account for contract multipliers, leverage, liquidity, and the selected price-move assumption. This expression is a sizing constraint, not a complete portfolio-risk model. If prices gap, liquidity disappears, or losses exceed the modeled scenario, realized losses may exceed the budget. Position size must also satisfy independent notional, leverage, concentration, and liquidity limits.

B. Volatility Targeting and Conservative Scaling

Where return estimates and volatility forecasts are sufficiently reliable, a strategy can be scaled inversely with estimated risk. Let Let σ_{target} denote the chosen volatility target and $\hat{\sigma}_t$ the forecast volatility at time t. A simplified scaling coefficient is defined by Equation (7).

$$w_t = \min(w_{\max}, \frac{\sigma_{\text{target}}}{\hat{\sigma}_t}) \tag{7}$$

The coefficient w_t is capped by the maximum permitted exposure $w_{\max}$. This approximation requires a strictly positive volatility estimate and an independently enforced position limit. Practical implementations must also handle forecast instability, estimation windows, volatility shocks, correlation changes, and liquidity constraints. Scaling should be reduced when model uncertainty or market disorder increases. Volatility targeting does not eliminate tail risk, and leverage must never be increased automatically beyond approved limits.

C. Portfolio-Level Constraints

Risk cannot be assessed by examining each strategy independently. Several strategies may share instruments, factors, venues, or correlated sources of liquidity, creating hidden concentration. Aggregate exposures across open orders and existing positions, include pending cancellations until their status is confirmed, and account for cross-strategy dependencies. Stress scenarios should include sharp price moves, widening spreads, funding changes, correlated drawdowns, exchange disconnections, and simultaneous liquidation of positions. Where correlation estimates become unstable, use conservative concentration caps rather than assuming historical diversification will persist.

D. Independent Pre-Trade and Post-Trade Controls

Strategy logic must not have unilateral authority over every risk boundary. An independent control layer validates proposed orders before submission and reconciles observed executions afterward. Pre-trade checks should reject orders that exceed approved quantity, notional, price-deviation, leverage, or aggregate-exposure limits. Post-trade checks should reconcile fills, positions, balances, realized PnL, and open-order state against the authoritative venue record. The SEC's Rule 15c3-5 establishes specific market-access control obligations for covered broker-dealers, including controls designed to limit financial exposure and prevent erroneous orders [6]. Its precise legal application depends on the entity and activity; the engineering principle of independent limits remains relevant more broadly.

E. Failure Detection and Emergency Stops

Define measurable triggers for abnormal operation before deployment. Examples include stale market data, rejected-order bursts, missing acknowledgements, duplicate submissions, unexpectedly large slippage, unexplained position discrepancies, breached loss limits, and repeated connectivity failures. A triggered stop must block new risk-increasing orders and follow a documented process for handling existing orders and positions. Cancelling open orders does not guarantee that exposure disappears immediately, because cancellation acknowledgements and fills may race. Recovery requires reconciliation, an explicit reset condition, and authorized approval rather than an automatic restart.

Table V summarizes the core controls and their required responses. Thresholds must be calibrated to the venue, instruments, strategy, and governing obligations; the table does not prescribe universal numerical limits.

TABLE V
Operational Risk Controls and Required Responses

Risk control	Detection condition	Required response
Exposure limits	Quantity or notional breach	Reject new order
Price validation	Erroneous price or size	Block submission
Data integrity	Stale or inconsistent feed	Suspend new decisions
Execution quality	Slippage exceeds threshold	Reduce risk or halt
Position reconciliation	Internal and venue states differ	Freeze and reconcile
Loss limits	Daily or portfolio limit breached	Invoke approved stop

F. Paper Trading and Staged Promotion

After passing research validation, each strategy enters a forward paper-trading phase using live or appropriately delayed market observations and a clearly labeled simulated execution model. Compare its expected decisions with timestamped signals, hypothetical fills, estimated costs, and realized market conditions. Document differences between simulated and attainable execution. Promotion requires prespecified evidence of stable behavior, acceptable risk, operational reliability, and satisfactory reconciliation. Capital allocation begins conservatively and increases only through an explicit review process. Any material strategy change resets the relevant validation requirements.

G. Governance and Auditability

Record strategy versions, parameter configurations, data provenance, approvals, risk thresholds, alerts, order events, stop activations, and manual interventions in an immutable or tamper-evident audit trail. Every promotion and emergency reset should have a recorded rationale and authorized owner. Separate strategy development from approval of risk limits wherever practical. Automated systems should fail closed when critical risk state stste cannot be verified. These controls turn deployment from an informal judgment into a repo-ducible engineering decision and establish the operational evidence needed for the final research protocol.

VI. Continuous Research, Limitations, and Conclusion

A reverse-engineering investigation should not terminate when an independent implementation matches selected observations. The resulting model must remain reproducible, responsive to new evidence, and open to rejection when market behavior changes. This final section defines a disciplined research loop that connects software investigation, quantitative evaluation, and prospective monitoring without assuming that an observed institutional mechanism can be copied completely or monetized automatically.

A. The Continuous Investigation Loop

Each completed investigation produces a versioned hypothesis, its supporting evidence, an independent implementation, and a record of validation results. New observations can then test whether the reconstructed mechanism remains consistent with behavior that was not previously examined. A change in software version, order-handling behavior, market structure, or data quality should trigger a targeted re-investigation instead of silently updating the model. REA supports repeatable agent-assisted investigation where its conffgured tools and target compatibility permit; conventional testing and quantitative analysis remain necessary for assessing economic value [1].

Use a scheduled review to compare predictions with new observations, categorize discrepancies, and prioritize the next experiment. A useful discrepancy taxonomy includes data error, incorrect behavioral inference, implementation defect, regime change, and execution-model mismatch. Each category requires a different response. Data errors require correction and replay; an incorrect inference requires a revised hypothesis; an implementation defect requires a code fix and regression tests; a regime change requires renewed statistical analysis; and an execution mismatch requires a revised fill or cost model.

B. Prioritizing Research Under Limited Resources

Research time should be allocated according to uncertainty, potential information gain, and test cost rather than the apparent profitability of the current best candidate. Let ΔU_i denote the expected reduction in uncertainty from experiment i, V_i the research value of resolving that uncertainty, and C_i the estimated experiment cost. A simple prioritization score is defined in Equation (8).

$$P_i = V_i \cdot \Delta U_i / C_i \tag{8}$$

The score is a transparent scheduling heuristic, not a statistical probability or trading-performance measure. Its inputs must be defined consistently and should not be tuned to force a favored strategy to the top of the queue. Experiments with high expected information value may be valuable even when they are likely to disprove the current hypothesis. Record both successful and unsuccessful investigations so future work benefits from the information acquired.

C. From Gross Signal to Economic Feasibility

A statistically detectable pattern becomes economically relevant only if its expected benefit survives execution costs, capacity limits, operational expenses, and risk constraints. For a candidate strategy, define expected net return per unit of traded notional in Equation (9), with all terms measured on a consistent basis.

$$E[R_{\text{net}}] = E[R_{\text{gross}}] - E[C_{\text{execution}}] - E[C_{\text{funding}}] - E[C_{\text{operational}}] \tag{9}$$

The cost terms must be estimated from evidence appropriate to the instrument and execution method. Historical candle data may support rough sensitivity analysis but usually cannot identify queue position, fill probability, or market impact precisely. Capacity must also be investigated: as order size increases, available liquidity may be insufficient and execution impact may rise. A positive gross signal is not proof of positive net returns, and a positive historical net return is not sufficient evidence of future profitability.

D. Reproducibility and Research Governance

Every experiment should preserve a machine-readable record of the target and version, evidence identifiers, hypothesis, features, code revision, data interval, parameter settings, execution assumptions, test splits, performance metrics, and rejection criteria. Random seeds, dependencies, configuration files, and data checksums should be recorded whenever relevant. Results must be reproducible from retained inputs. Report all attempted variants and failed experiments because selective disclosure of the best outcome can materially overstate performance, particularly when many candidate strategies have been compared [2], [4].

Keep software reconstruction separate from live-trading authorization. A model that reproduces a documented fee calculation or observed order transition should first pass deterministic tests and independent review. A strategy derived from the model must then pass statistical validation, paper trading, risk approval, and controlled operational testing. No automated research process should be permitted to circumvent access controls, extract private credentials, or deploy a strategy simply because a backtest appears attractive.

E. Research Limitations

The framework has important limits. REA's results depend on the artifact, supported analysis provider, configured MCP tools, and quality of available evidence. Public observations rarely identify a unique internal strategy, while decompilation may recover pseudocode or structural clues rather than original source. Market data can reveal associations without proving causality. Backtests depend on assumptions about execution, sampling, costs, and strategy selection. No tool can guarantee recovery of confidential hedge fund logic, and no reconstruction method guarantees positive returns. These limits are properties of the evidence and identification problem, not defects that can be removed by adding more model complexity.

F. Conclusion

This paper presents a practical framework for investigating authorized software and observable trading behavior, converting findings into independent quantitative models, and evaluating those models with realistic costs, statistical controls, and operational safeguards. REA provides a structured route from investigation to reconstruction; quantitative research determines whether a resulting hypothesis explains the data and survives independent testing. The appropriate claim is therefore not that any hedge fund can be completely concracked. It is that a disciplined researcher can systematically examine accessible evidence, construct reproducible models of selected mechanics, and reject unsupported explanations. The ultimate deliverable is an auditable research process—not a promise of hidden alpha.

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